

AN **INDIAN** ADAPTATION

Third Edition

Supply Chain Management

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SUBJECT: SUPPLY CHAIN MANAGEMENT STRATEGY

CHAPTER: 08

CHAPTER TITLE: Forecasting and Demand Planning

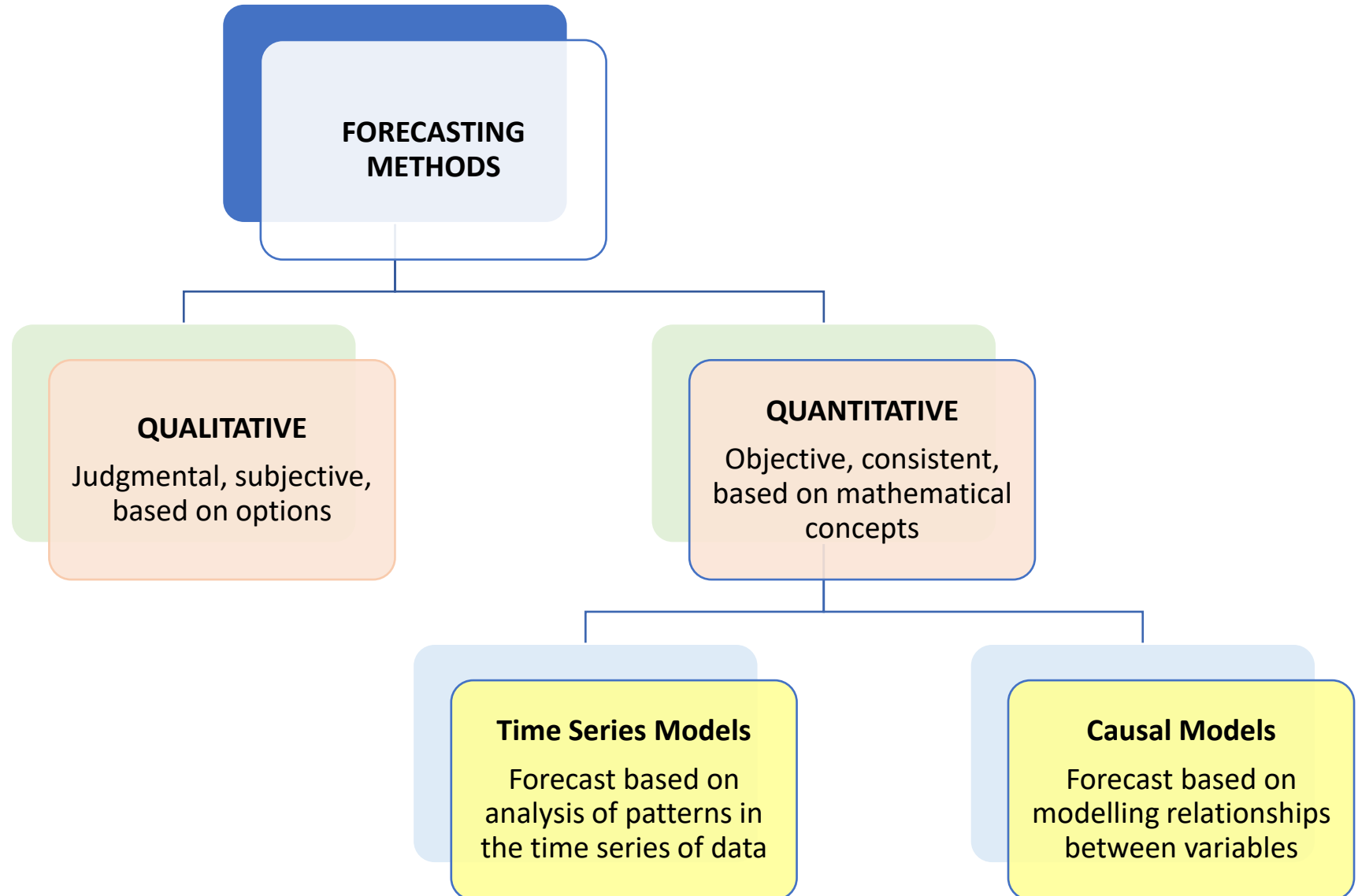
REFERENCE BOOK:

SUPPLY CHAIN MANAGEMENT (AN INDIAN ADAPTATION), N R SANDERS & S S PADHI, WILEY, 3RD ED. (2023)

Forecasting

- Forecasting is one of the most important business activities, as it drives all other business decisions.
- Decisions such as which markets to pursue, which products to produce, how much inventory to carry, and how many people to hire are based on a forecast.
- Forecasting is a challenge.
- Poor forecasts result in poor planning and can leave the company unprepared to meet future demands.
- The consequences can be costly in terms of lost sales or excess inventory that can not be sold.

Types of Forecasting Methods



Time Series Forecasting Models

- A time series is simply a listing of data points of the variable being forecast over time taken at regular intervals.
- For example: Census data over the past 50 years.

(1) Mean:

- The simplest forecasting models is the mean.

$$F_{t+1} = \frac{\sum D_t}{n}$$

or

$$F_{t+1} = \frac{[D_t + D_{t-1} + \dots + D_{t-n}]}{n}$$

Where, F_{t+1} = Forecast of demand for next period, t+1

D_t = Demand for current period, t

n = Number of data points

- A reasonable forecasting model to use for forecasting stable and mature products.

Time Series Forecasting Models

Question:

Handy power tool is forecasting sales of its popular drill. The company has been selling the drill for the past 15 years, and sales have been steady. The company uses a simple mean to forecast weekly sales. Sales over the past five weeks are available, and the company will use them to make a forecast for week six.

Answer:

The forecast is simply an average of the past five weeks:

$$F_{t+1} = \frac{[D_t + D_{t-1} + \dots + D_{t-n}]}{n}$$

$$\text{Or } F_6 = \left[\frac{8+10+9+12+10}{5} \right] = \frac{49}{5} = 9.8 = 10$$

Week	Drill Sales
1	8
2	10
3	9
4	12
5	10

Problem:

Suppose the demand for a product in the last 4 months (January-April) is given, then find out the demand for May?

Month	Demand
January	100
February	120
March	110
April	130

Time Series Forecasting Models

(2) Moving Average:

- It generates a forecast by averaging a specifies number, n , of the most recent data rather than the entire data set.
- As new data become available, the oldest are dropped, and the number of observations used to compute the average is kept constant.
- The simple moving average “moves” through time.

$$F_{t+1} = \frac{\sum D_t}{n}$$

or

$$F_{t+1} = \frac{[D_t + D_{t-1} + \cdots + D_{t-n}]}{n}$$

Where, n = Number of data points in the moving average

Time Series Forecasting Models

Question:

Ramesh forecasts sales of kitchen aprons using a **three-period moving average**. How many aprons should they expect to sell in April given the following sales figures from January to March? To find the forecast for April we take an **average of the last three observations**

Answer:

$$F_{t+1} = \sum D_t / n$$

$$\text{Or } F_{\text{April}} = \left[\frac{38+27+42}{3} \right]$$
$$= 35.7 \approx 36 \text{ aprons.}$$

Problem:

Suppose the demand for a product in different months (January-March) is given. How many products should they expect to sell in April? Use two-period moving average.

Month	Demand
January	50
February	60
March	70

Causal Models

Linear Regression:

- It assumes a linear, or straight line, relationship between two variables.
- The variable being forecast, called the dependent variable, is linearly related to another variable, called the independent variable.
- **When** we fit the data, the line must be a straight line.

$$Y = a + bX$$

Where, Y= Dependent variable,

X= Independent variable,

a = Y intercept of the straight line,

b= slope of the straight line

Compute Parameter

- Using the linear regression requires first developing the linear regression equation.
- Any value of the independent variable, we can compute the value of the dependent variable, which is the variable being forecast.

Step-1: Compute parameter b

$$b = \frac{n \sum XY - (\sum X)(\sum Y)}{n \sum X^2 - (\sum X)^2} \quad OR \quad b = \frac{\sum XY - n\bar{X}\bar{Y}}{\sum X^2 - n\bar{X}^2}$$

Where, $\bar{Y}$ = average of the Y values

$\bar{X}$ = average of the X values

n = number of data points

Compute Parameter

Step 2: Compute parameter a

$$a = \bar{Y} - b\bar{X} \quad \text{or} \quad a = \frac{\sum Y - b \sum X}{n}$$

Step 3: Substitute values for a and b in the equation

$$Y = a + bX$$

Step 4: Generate a forecast for the dependent variable (Y)

Substitute the appropriate value for the independent variable(X).

Example of Forecasting with Linear Regression

Question:

Pizza boy is trying to forecast the volume of pizza sold based on the amount of money it has put into advertising in the local newspaper. It has been tracking the relationship between pizza sales and advertising over the past four months. The results are as follow:

Pizza Sales (Y)	Advertising (Dollars) (X)
58	135
43	90
62	145
68	145

What would happen to pizza sales if Pizza boy invested \$150 in advertising for next month?

Example of Forecasting with Linear Regression

Answer:

In this example pizza sales are the dependent variable (Y) and advertising dollars the independent variable (X).

	Pizza Sales (Y)	Advertising (Dollars) (X)	Y^2	X^2	XY
	58	135	3364	18225	7830
	43	90	1849	8100	3870
	62	145	3844	21025	8990
	68	145	4624	21025	9860
SUM	231	515	13681	68375	30550

$$\bar{X} = \frac{515}{4} = 128.75 \text{ and } \bar{Y} = \frac{231}{4} = 57.75$$

Step 1 Compute parameter b:

$$b = \frac{\sum XY - n\bar{X}\bar{Y}}{\sum X^2 - n\bar{X}^2} = \frac{30550 - 4(128.75)(57.75)}{68375 - 4(128.75)^2} = 0.391$$

Step 2 Compute parameter a:

$$a = \bar{Y} - b\bar{X} = 57.75 - (0.391)128.75 = 7.41$$

Step 3 Substitute values for a and b in the equation:

$$Y = a + bX = 7.41 + 0.391X$$

Step 4 Generate for the dependent variable (Y):

Substitute the appropriate value for the independent variable (X).

$$Y = 7.41 + 0.391(150) = 66 \text{ pizzas}$$

Example of Forecasting with Linear Regression

Problem:

A coffee shop wants to forecast monthly sales based on advertising expenditure. What will be the expected sales if the shop spends Rs. 60 on advertising.

Advertising (Rs)	Sales (Thousand)
20	40
30	50
40	65
50	70

Measuring Forecast Accuracy

- Measuring forecast accuracy is a critical aspect of forecasting.
- Data can change over time, and models that once provided good results may no longer be adequate.
- The model's accuracy can be assessed only if forecast performance is measured over time.
- The first step in measuring forecast accuracy is to measure the forecast error.
- Forecast error is the difference between actual demand and the forecast for a given period:

$$e_t = D_t - F_t$$

Where, e_t = Forecast error for period t

D_t = Actual demand for period t

F_t = Forecast for period t

Measuring Forecast Accuracy

- However, computing error for one time period does not provide enough information.
- Two of the most commonly used error measures are the Mean Absolute Deviation (MAD) and the Mean Square Error (MSE).
- MAD is the average of the sum of the absolute error:

$$\text{MAD} = \frac{\sum |Actual - Forecast|}{n}$$

- MSE is the average of the squared error:

$$\text{MSE} = \frac{\sum (Actual - Forecast)^2}{n}$$

Measuring Forecast Accuracy

Question:

Harry Thurber Apparel is comparing the accuracy of two methods that it has used to forecast monthly sales of its popular jeans. Forecasts of monthly sales from January through May for method A and method B are shown below against the actual sales. Which method provided better forecast accuracy?

Month	Sales	Method A				Method B			
		Forecast	Error	Error	Error ²	Forecast	Error	Error	Error ²
Jan	40	42	-2	2	4	44	-4	4	16
Feb	28	29	1	1	1	31	-3	3	9
Mar	41	39	2	2	4	38	3	3	9
Apr	41	38	3	3	9	42	-1	1	1
May	39	41	-2	2	4	40	-1	1	1
Total			2	10	22		-6	12	36

Accuracy for method A

$$\text{MAD} = \frac{\sum |\text{Actual} - \text{forecast}|}{n} = \frac{10}{5} = 2$$

$$\text{MSE} = \frac{\sum (\text{Actual} - \text{Forecast})^2}{n} = \frac{22}{5} = 4.4$$

Accuracy for method B

$$\text{MAD} = \frac{\sum |\text{Actual} - \text{forecast}|}{n} = \frac{12}{5} = 2.4$$

$$\text{MSE} = \frac{\sum (\text{Actual} - \text{Forecast})^2}{n} = \frac{36}{5} = 7.2$$

Of the two methods, method A produced both a lower MAD and a lower MSE, which means that it provided better forecast accuracy.

Measuring Forecast Accuracy

Problem:

- A company forecasted sales for 6 months. The actual and forecasted sales (in units) are given below. Using Mean Absolute Deviation (MAD) and Mean Squared Error (MSE), analyse and comment on the forecast accuracy

Period	Actual Demand	Forecast (A)	Forecast (B)
1	120	115	118
2	135	140	130
3	150	148	152
4	160	155	165
5	170	175	168
6	180	178	182

Thank You...